

Resistance to Doing Right: rising cost of misconduct from body-worn cameras adoption and its unintended consequences for de-policing and crime in vulnerable areas

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Abstract

This study investigates the impact of body-worn cameras (BWCs) on police behavior in São Paulo, Brazil, focusing on *de-policing*: a decline in proactive policing in response to criticism or the threat of punishment. Although BWCs have been widely studied for their effects on police violence and accountability, their unintended consequences on police retreat and consequently crime rates remain underexplored, especially in Brazil. Using a quasi-experiment methodology, we find strong evidence of *de-policing* in areas where BWCs were implemented, accompanied by a notable increase in crime. These findings address a critical gap in the literature and highlight the complex and unexpected effects of BWC adoption.

Keywords: Body-worn camera, De-policing, Police brutality, Body cam, Crime

1. Introduction

The implementation of body-worn cameras (BWCs) in policing has become an internationally recognized practice to curb abuses in police interventions, widely seen as an effective tool to enhance transparency, protect civilians, and ensure accountability in officers' activities. Empirical studies have documented substantial reductions, following the introduction of BWCs, in officer firearm usage (Ferrazares, 2024) and deaths from police interventions (Tavares, Marcolino, & Cabral, 2024), aligned with the evidence of their effectiveness (Lum et al., 2019). However, as with many public policies, unintended consequences may also emerge. Recently, the international literature on police activity has drawn attention to the so-called *Ferguson Effect* — the observed reductions in police activity following intense public scrutiny or criticism after high-profile incidents of police violence, as officers become more cautious or reluctant to engage proactively (Shjarback et al., 2017). In particular, BWCs may lead to reduced police activity — known as *de-policing* (Nix & Wolfe, Dec. 2016) — as officers deterred by oversight may choose inaction over engagement.

This potential decline in police activity — driven by a behavioral shift from misconduct to inaction, rather than to technical intervention — can be attributed to several factors: a lack of incentives for proactive and technical policing, backlash against institutional oversight and deliberate boycotts, officers' fear of misinterpretation when acting in a vague context, additional bureaucratic workload associated with camera use, insufficient training for high-stress or complex situations, among others. These challenges are especially pronounced in contexts with historically weak oversight and where non-technical or violent practices have become normalized forms of police engagement, as is often the case in vulnerable Brazilian communities (Magaloni, Melo, & Robles, 2023). In turn, a reduction in police engagement, particularly in low-income and peripheral areas, may inadvertently contribute to rising local crime rates (Shi, 2009).

Although evidence that BWCs can reduce police violence is well established, their effects on *de-policing* and crime remain limited or unexplored. This article theoretically examines police behavior from a rational choice perspective, drawing on Becker (1968) to associate costs and benefits with non-technical conduct, technical engagement, and police inaction. Assuming officers behave rationally, we hypothesize about how BWCs alter the costs of misconduct and what conditions are required for this behavioral shift to improve policing rather than discourage it.

We investigate these dynamics using the case of São Paulo, Brazil, where the state government implemented BWCs across multiple police battalions. Employing a quasi-experimental difference-in-differences approach and confidential administrative data from the São Paulo Military Police from 2020 to 2023, we assess whether the adoption of BWC is associated with a decline in police confrontations — officers' interventions resulting in imprisonment, death, or injury — and the subsequent effects on property crimes such as mobile phone and vehicle theft. Our findings reveal that BWCs contributed to a measurable decrease, particularly in lower-income areas, in police confrontations. Even though it remains uncertain whether this reduction reflects improved effective officer action or *de-policing*, we find a significant increase in property crime in areas where BWCs were implemented, consistent with the *de-policing* hypothesis. These results underscore that even well-intentioned accountability reforms can produce unintended consequences when applied in specific institutional contexts.

This study advances the debate on body-worn cameras by addressing a critical gap in understanding their broader behavioral and public safety effects in developing contexts. While most research has focused on the capacity of BWCs to reduce police violence, much less is known about how they reshape officer behavior and local crime dynamics, particularly in socially unequal and institutionally fragile environments such as Brazil. By examining São Paulo’s large-scale BWC implementation, this paper highlights how accountability reforms can generate uneven patterns of police engagement and unintended consequences for community safety. These insights contribute to a growing literature on the political economy of police reform, emphasizing the importance of implementation design and local context in ensuring that transparency efforts strengthen, rather than weaken, state presence in vulnerable areas.

2. Context

Brazil has long dealt with significant challenges related to police violence and racial disparities in law enforcement. Studies consistently show that the country experiences some of the highest rates of police lethality in the world, disproportionately affecting black and low-income populations (Barbosa et al., 2025). National-level data from the Brazilian Forum on Public Security indicate that police officers killed over 6,000 people in 2024 alone, and 82% of those were Black (Fórum Brasileiro de Segurança Pública, 2025). These patterns reflect deep-rooted structural inequalities and a historical reliance on militarized police, which have often prioritized aggressive tactics in marginalized communities (Magaloni, Melo, & Robles, 2023).

Between 2014 and 2024, the proportion of violent deaths caused by police interventions in Brazil rose sharply, increasing from 5.1% to 14.1% (Fórum Brasileiro de Segurança Pública, 2025). Data from the São Paulo State Public Security Department indicate that, although the state has experienced overall declines in homicide rates in recent decades, deaths resulting from police interventions remain high and accounted for 22% of all violent deaths in São Paulo in 2024, a figure above the national average (Fórum Brasileiro de Segurança Pública, 2025).

In this setting, the introduction of BWCs in São Paulo’s Military Police represented a significant development in efforts to address police accountability and curb excessive use of force. The implementation has been phased, starting in June 2021, and by March 2023, 10,125 cameras had been deployed across sixty-four battalions in the state (Carvalho & Rossano, 2023). The BWCs are designed for frontline policing, recording the entirety of the 12-hour shift. The equipment operates with two modes: a continuous “video-recall” (routine) mode, which records without sound at 480 pixels for privacy, and an “intentional recording” mode, which, when activated manually, records with sound at 720 pixels, provides GPS and live streaming, and retroactively captures the preceding 90 seconds. Officers are unable to manually disable the continuous recording, and detailed protocols clearly delineate the circumstances for initiating intentional recording, such as during the use of force or police stops. Furthermore, officers must, when safe and possible, verbally inform citizens that they are being recorded, but consent is not required (Carvalho & Rossano, 2023). Unlike some jurisdictions where camera adoption is voluntary, São Paulo mandated the use of BWCs across multiple battalions, minimizing self-selection biases and strengthening the impact of the program on police behavior.

The adoption of body-worn cameras became a salient issue in São Paulo’s 2022 state election, reflecting broader ideological divides over public security and police accountability. The election of Tarcísio de Freitas, a right-wing candidate running on a conservative, tough-on-crime platform, brought renewed scrutiny to the BWC program. While the candidate did not reject the use of cameras outright, his campaign discourse emphasized concerns about excessive external oversight, potential constraints on police discretion, and the impact of continuous recording on officers’ ability to make rapid decisions in high-risk encounters.

While the initial rollout required continuous recording to prevent selective activation and ensure comprehensive oversight, Tarcísio’s proposal pushed for greater reliance on manual activation by officers or the possibility of remote deactivation by central command. Human rights advocates argue that these changes risk undermining transparency and weakening the accountability gains achieved during the early phases of implementation.

Our analysis focuses on evaluating the effect of the introduction and adoption of Body-Worn Cameras in the São Paulo State Military Police during the period preceding the inauguration of Governor Tarcísio de Freitas in January 2023. We investigate whether the implementation of BWCs in São Paulo has had the unintended effect of reducing policing activity and explore how these dynamics intersect with broader public safety outcomes. Therefore, the conclusions and variables analyzed represent the original implementation scenario, before any potential public policy changes by the Tarcísio administration could influence the measured indicators.

3. Empirical literature review

A growing empirical literature has examined the effects of body-worn cameras (BWCs) as accountability devices aimed at curbing police misconduct, excessive use of force, and abusive practices. Early studies, largely conducted in the United States, focused on whether BWCs alter officer behavior in police–citizen encounters by increasing transparency and the expected cost of misconduct. Randomized controlled trials and quasi-experimental evaluations generally find that BWCs are associated with reductions in use-of-force incidents and civilian complaints. For example, Braga et al. (2018) report that

officers equipped with cameras generate fewer complaints and use-of-force reports relative to their non-equipped counterparts, while continuing to engage actively with the public. Similarly, [Headley, Guerette, & Shariati \(2017\)](#) find that BWCs lead officers to rely on less intrusive methods of resolving incidents without evidence of widespread disengagement. Using administrative data from Chicago, [Ferrazares \(2024\)](#) shows that BWC adoption is associated with substantial declines in use-of-force complaints and officer-reported physical force, particularly in racially asymmetric interactions, suggesting that cameras meaningfully constrain aggressive and potentially unlawful behavior. This aligns with broader economic models of policing, which suggest that increased monitoring—whether internal or external—can significantly alter officer behavior and reduce misconduct allegations ([Owens & Ba, 2021](#)); ([Rivera & Ba, 2026](#)); ([Rozema & Schanzenbach, 2023](#)). Taken together, this body of evidence supports the view that BWCs can function as effective monitoring technologies, improving compliance with procedural norms and reducing overt manifestations of police misconduct, even though results can vary based on discretion and compliance levels ([Lum et al., 2019](#)).

While the majority of BWC evaluations have been conducted in the United States, recent scholarship has examined the effects of this technology in the distinct context of developing countries, particularly in Brazil, where rates of police lethality are significantly higher. In one of the first randomized field experiments conducted in a high-violence setting in the Global South, [Magaloni, Melo, & Robles \(2023\)](#) study the deployment of BWCs in Rio de Janeiro’s Rocinha favela. They find that camera assignment leads to sizable reductions in stop-and-search activity, aggressive interactions, and other potentially abusive practices, even in a context characterized by militarized policing and weak accountability. Complementary evidence from Brazil is provided by [Tavares, Marcolino, & Cabral \(2024\)](#), who document a significant decline in deaths resulting from police intervention following the adoption of BWCs by the São Paulo Military Police. Their findings suggest that part of this reduction may operate through decreased police engagement in high-risk confrontational situations. More recently, [Barbosa et al. \(2025\)](#), using a randomized experiment in Santa Catarina, show that BWCs dramatically reduce police use of force and negative interactions while improving reporting quality, providing some of the strongest causal evidence to date from a developing-country setting. Together, these studies indicate that BWCs can substantially restrain violent and authoritarian policing in contexts where baseline levels of abuse are high, while also raising questions about how changes in police behavior propagate beyond misconduct reduction.

Parallel to the literature on BWCs, a distinct but related body of research has examined the phenomenon of de-policing and the so-called Ferguson Effect. De-policing refers to a withdrawal from proactive law enforcement activities by officers in response to heightened scrutiny, criticism, or perceived legal risk, while the Ferguson Effect specifically links such behavioral changes to high-profile incidents of police violence and subsequent public backlash. Empirical studies provide robust evidence that salient episodes of police violence can induce sharp declines in discretionary policing. Using detailed administrative data, [Shjarback et al. \(2017\)](#) document substantial reductions in stops, searches, and arrests across Missouri police departments following the Ferguson protests, with particularly pronounced declines in jurisdictions serving larger Black populations. [Cheng & Long \(2022\)](#) offer direct causal evidence that highly publicized police killings lead to significant reductions in self-initiated policing activities and arrests in large U.S. cities, consistent with officers retreating from discretionary enforcement. [Kim \(2024\)](#) similarly shows that proactive policing in New York City fell sharply in 2020 following the COVID-19 pandemic and the killing of George Floyd, with heterogeneous effects across neighborhoods. Importantly, these studies demonstrate that de-policing can arise even in the absence of formal policy changes, driven instead by shifts in perceived oversight, legitimacy, and risk.

The relationship between de-policing and crime, however, remains contested. While some narratives attribute post-Ferguson increases in violent crime to police withdrawal, empirical findings are mixed. [Rosenfeld & Wallman \(2019\)](#) find no evidence that declining arrest rates caused the spike in homicide observed in large U.S. cities in 2015, suggesting that de-policing alone cannot explain crime dynamics. In contrast, [Cheng & Long \(2022\)](#) report localized increases in violent crime following highly publicized police killings, highlighting the possibility that reductions in police activity and deteriorating community cooperation may jointly influence public safety outcomes. Complementary work by [Ang et al. \(2025\)](#) shows that high-profile acts of police violence substantially reduce community engagement with law enforcement, as measured by declines in 911 calls relative to gunshot detection, underscoring the role of civilian behavior and trust in mediating the effects of police withdrawal. Together, this literature emphasizes that de-policing is a multifaceted phenomenon, shaped by both officer and civilian responses to heightened scrutiny.

Only recently has research begun to connect these two strands by examining whether BWCs themselves may induce de-policing or alter crime outcomes. This emerging literature suggests that while BWCs reduce misconduct, they may also change officers’ cost-benefit calculus in ways that discourage proactive engagement. [Wallace, Katz, & Choate \(2018\)](#), analyzing a randomized rollout of BWCs in Spokane, test the hypothesis of “camera-induced passivity” and find little evidence of reduced officer activity on average, though they emphasize that effects may depend on organizational context. [Bennett et al. \(2023\)](#) similarly find no reduction in foot patrols or traffic stops associated with BWCs in a large U.S. department. In contrast, [Ferrazares \(2024\)](#) documents a significant decline in drug-related arrests following BWC adoption in Chicago, consistent with selective de-policing in discretionary enforcement domains. Evidence from developing contexts appears even more suggestive: [Magaloni, Melo, & Robles \(2023\)](#) find that BWCs not only reduce aggressive encounters but also substantially decrease overall police activity, including responses to citizen requests, unless supervisory monitoring is simultaneously strengthened.

Despite these advances, the empirical literature on BWCs, de-policing, and crime remains fragmented and incomplete. Existing studies often focus on single outcomes, short time horizons, or specific institutional settings, and relatively few examine downstream effects on local crime, particularly in vulnerable or high-violence areas. Moreover, most evidence comes from the United States, limiting external validity for contexts characterized by weaker oversight, higher baseline misconduct, and greater reliance on coercive policing. As a result, while BWCs are widely promoted as tools to enhance accountability, much less is known about their broader behavioral consequences and potential trade-offs for public safety. This gap is especially salient in developing-country settings, where reductions in abusive policing may coexist with declines in proactive enforcement. By systematically examining how BWCs reshape police behavior and crime dynamics across heterogeneous urban areas, further empirical research can help clarify when accountability reforms strengthen policing and when they risk producing unintended security vacuums.

4. Theoretical framework and hypotheses

This study draws on [Becker \(1968\)](#)’s economic theory of crime, which posits that individuals engage in criminal behavior when the expected utility of the act exceeds its expected cost, considering the likelihood of detection and the severity of punishment. While originally developed to explain criminal decision-making, this framework has been increasingly applied to understand discretionary behavior within institutional contexts — particularly in policing, as in [Cheng & Long \(2022\)](#) and [Rozema & Schanzenbach \(2023\)](#). From this perspective, a police officer’s conduct in the field can be modeled as the outcome of a cost-benefit analysis among three primary alternatives: non-technical policing (misconduct or excessive force, which is illegal), technical policing (procedurally correct action), and inaction (de-policing). Each choice involves trade-offs.

Officers may choose to act non-technically — for instance, by using excessive force or procedural shortcuts — because such behavior may provide immediate non-pecuniary rewards, such as expedience, a perceived sense of control over volatile situations, or informal peer approval in organizational environments where aggressive policing is culturally legitimized. In police institutional contexts characterized by a “warrior” or punitive ethos — often summarized in slogans such as “a good criminal is a dead criminal” — the use of force can be tacitly encouraged, normalized, and even celebrated among colleagues. Such approval may extend beyond horizontal peer recognition to vertical endorsement by supervisors, who may reward officers perceived as “tough” or “effective” through informal protection, desirable assignments, or career advancement opportunities over the medium and long run. However, non-technical conduct also carries expected costs related to the probability of detection, the likelihood of punishment if detected, and the severity of the sanction.

Officers may also choose to engage in technical policing, which complies with formal institutional norms and legal standards. We assume that pecuniary returns (e.g., salary) remain constant across choices, since the police administration in São Paulo currently lacks structured incentive programs that reward productivity, the quality of interactions, or local crime outcomes. Nevertheless, technical policing may generate non-pecuniary benefits such as professional satisfaction, reputational gains, and adherence to ethical or legal duty — but these benefits are conditional on an institutional environment that recognizes, values, and protects procedurally correct conduct. In organizational contexts where such recognition is weak or absent, the non-pecuniary returns to technical policing may be limited or uncertain. Moreover, even when present, they may be offset by higher perceived effort, greater exposure to physical risk, fear of misinterpretation (especially under surveillance), and increased bureaucratic burden.

As pecuniary benefits are assumed to be constant across options, inaction may offer limited non-pecuniary rewards but also entails comparatively low expected costs. In some contexts, it may represent the least risky alternative, particularly when both forms of active engagement carry high expected costs. A key non-pecuniary advantage of inaction is the avoidance of physical danger, such as injury or death in confrontational encounters.

We can represent the expected utility of each option as U_j , where $j \in \{N, T, I\}$ stands for non-technical policing, technical policing, and inaction, respectively. Let B^P denote pecuniary benefits (e.g., salary, assumed constant across choices), B_j^N denote non-pecuniary benefits specific to each choice, C_j the cost of each choice j , and p_d and p_p the probabilities of misconduct detection and punishment, respectively. When $j = N$, the expected cost reflects the probability-weighted cost of punishment for misconduct. Then:

$$\begin{aligned} U_N &= B^P + B_N^N - (p_d \cdot p_p \cdot C_N) \\ U_T &= B^P + B_T^N - C_T \\ U_I &= B^P + B_I^N - C_I \end{aligned}$$

The officer chooses the action that maximizes expected utility:

$$\text{Decision} = \arg \max_{j \in \{N, T, I\}} U_j$$

The introduction of BWCs fundamentally shifts this cost-benefit calculus by increasing the expected cost of non-technical policing. Cameras increase the probability of detection (p_d), often the probability of punishment conditional on

detection (p_p), and the expected severity of the sanction (C_N) by producing objective audiovisual evidence. Consequently, the expected utility of misconduct (U_N) falls, discouraging non-technical conduct¹.

However, this shift does not necessarily induce officers to substitute non-technical behavior with procedurally correct conduct. Instead, they may transition from misconduct to inaction when the expected utility of not acting exceeds that of acting technically ($U_I > U_T$) and the cost-benefit balance of inaction becomes more favorable than engagement.

At the same time, BWCs may unintentionally increase the perceived cost of technical policing, particularly when implementation or oversight is flawed. Officers may fear that procedurally correct conduct could be misinterpreted or judged out of context, especially in high-stress or ambiguous encounters. Additional bureaucratic burdens — such as expanded reporting requirements or camera management — can further discourage proactive engagement. Moreover, insufficient training on how to operate effectively under continuous surveillance may cause officers to hesitate or disengage altogether. Formally, when the perceived cost of technical action also rises, the utility of the technical policing decreases ($\Delta C_T > 0 \Rightarrow \Delta U_T < 0$), reducing the attractiveness of procedurally correct conduct relative to inaction. If the introduction of BWCs substantially increases the cost of non-technical policing and, to a lesser extent, raises the cost of technical actions, inaction becomes the most rational choice. This mechanism is particularly salient in high-risk situations, such as armed confrontations with organized criminal groups.

For officers to transition from non-technical to technical conduct, the relative attractiveness of technical policing must increase — that is, the expected cost of acting correctly must fall, or its expected benefit must rise. Such changes can be achieved through institutional mechanisms, including performance-based reward programs (increasing pecuniary returns B^P specifically for technical conduct), organizational protection for correct decisions made under pressure (reducing C_T), or sanctions for unjustified inaction (effectively lowering the pecuniary payoff associated with inaction). Formally, officers will prefer technical over non-technical or inactive conduct when

$$U_T > \max(U_N, U_I),$$

which can occur if

$$\Delta B_T^P > 0, \quad \Delta C_T < 0, \quad \text{and/or} \quad \Delta B_I^P < 0.$$

This mechanism helps explain why, in practice, BWCs may lead not to better policing but to less policing — especially under poor implementation conditions.

Because policing decisions are highly context-dependent, the effects of BWCs are unlikely to be spatially uniform. Officers already tend to act more technically — and thus avoid misconduct — in wealthier, more central, and more closely monitored areas, where surveillance infrastructure and social accountability are stronger. In contrast, interactions with low-income and Black residents more often occur in environments with limited visibility and weaker accountability, where officers perceive lower probabilities of detection and lower expected sanction costs.

That is:

- $p_d^L < p_d^H$: the probability of detecting non-technical policing is lower in low-income areas.
- $C_N^L < C_N^H$: the expected punishment cost of non-technical policing is lower in low-income areas.

Here, the index $a \in L, H$ denotes the area type, distinguishing between low-income (L) and high-income (H) neighborhoods. From this, it follows that the expected utility of non-technical policing is higher in low-income areas ($U_N^L > U_N^H$), since the expected costs of detection and punishment are lower in these contexts.

Given this asymmetry, BWCs are likely to be more effective at deterring misconduct in low-income and vulnerable neighborhoods by introducing an accountability mechanism that is otherwise weaker in these settings. However, if the adoption of cameras does not increase the benefits or reduce the costs of technical policing, or raise the cost of inaction in these areas, officers may still rationally choose disengagement over engagement. This implies that BWCs may reduce misconduct more effectively in low-income neighborhoods while simultaneously triggering higher levels of de-policing there. Such an asymmetric incentive structure can generate selective disengagement, thereby reinforcing existing inequalities in public safety and state presence.

Extending back the Beckerian framework to the criminal side of the interaction helps explain the potential public safety implications of de-policing. When police activity declines, the expected probability of detection faced by potential offenders also decreases. As in Becker’s model, when the criminal’s perceived likelihood of apprehension and punishment falls, the expected utility of committing a crime rises, leading rational offenders to increase their criminal activity. In this sense, de-policing can indirectly exert upward pressure on local crime rates. However, this effect is unlikely to be immediate: it should emerge only over time, as potential offenders gradually perceive a sustained reduction in police presence and update their behavior accordingly.

Based on this framework, we propose the following hypotheses to be empirically tested:

¹This perspective is also grounded in deterrence theory, which posits that the constant risk of being recorded increases the perceived certainty of apprehension and punishment for misconduct. For both police officers and citizens, the camera might act as a source of formal surveillance, raising the potential costs of inappropriate or illegal behavior and thereby deterring such actions (Braga et al., 2018)

- H1: The adoption of body-worn cameras leads to a reduction in observed police activity, reflecting a retreat from non-technical policing that is not replaced by procedurally correct engagement, but rather by inaction (de-policing).
- H2: The reduction in police activity is more pronounced in low-income areas, where the expected cost of non-technical policing increases more sharply relative to high-income areas, while the incentives for technical engagement remain weak in both contexts.
- H3: The decline in proactive policing is followed by an increase in property crime, particularly in areas experiencing greater police withdrawal.

5. Data

We use three sets of data. First, about the policy implementation: when and where police officers started using body-worn cameras. Second, about police activity, based on records from proactive and reactive actions registered by officers. Lastly, about property crimes, based on geocoded official reports on cell phone and vehicle thefts.

5.1. Policy implementation

The implementation of body-worn cameras across police units in São Paulo was carried out in three groups of battalions over the following commands: Command of Policing of the Capital (CPC), the Command of Specialized Police Forces (CPChq), the Command of Policing of the Interior (CPI), and the Command of Policing of the Metropolitan Area (CPM). The separation of battalions into groups did not represent waves of policy implementation, as they overlapped in time, as 1 shows. Group 1 started with BWC in June 2021; Group 2 followed in early 2022; and, finally, Group 3 launched in mid-2022.

Group	Command	Implementation Dates
1st Group	CPC	2021-06-01, 2022-01-20
1st Group	CPChq	2021-06-01
1st Group	CPI	2021-06-01, 2022-01-20
1st Group	CPM	2021-06-01, 2022-01-20, 2022-08-31
2nd Group	CPC	2022-01-20, 2022-04-01, 2022-04-19, 2022-08-01
2nd Group	CPChq	2021-11-01
2nd Group	CPM	2022-04-01, 2022-04-19
3rd Group	CPC	2022-08-01, 2022-08-31
3rd Group	CPM	2022-08-01, 2022-08-31

Table 1: Camera Implementation Dates by Group and Command

Across groups, there are five cohorts of implementation: June 2021, November 2021, January 2022, April 2022, and August 2022, as represented in the maps of Figure 1². This staggered deployment offered the opportunity to adopt a research design to identify the causal impact of BWC on our variables of interest, as detailed in the section 6.

²The data of police confrontation allowed us to reconstruct the geographic boundaries of police battalions (*batalhões policiais*), a critical step given that the official delimitation of these areas is not publicly disclosed. Using the spatial distribution of police-initiated events, we apply clustering and density analysis to approximate the operational regions for each battalion. First, we extracted the coordinates of all geocoded occurrences and identified each event’s nearest neighbors. We then reassigned each point to the most common battalion label among its neighbors to reduce classification noise. Using the cleaned point cloud, we generated Voronoi polygons to estimate territorial divisions and dissolved them by dominant battalion assignment. Finally, we clipped the resulting polygons to the administrative boundaries of São Paulo state. This method produces contiguous, non-overlapping estimates of operational battalion areas, allowing for spatial aggregation and treatment assignment in subsequent analysis.

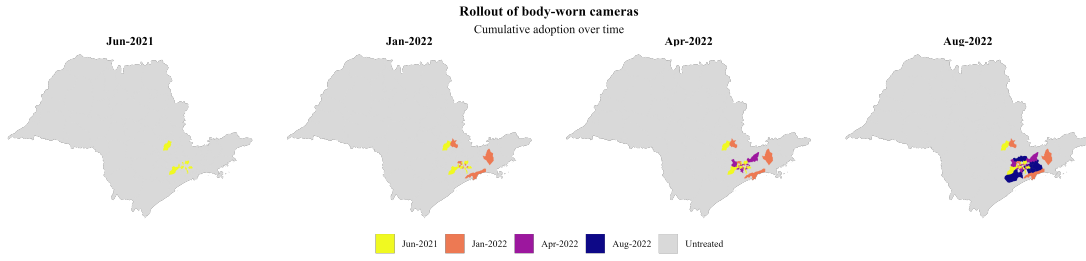


Figure 1: Temporal evolution of body-worn camera implementation.

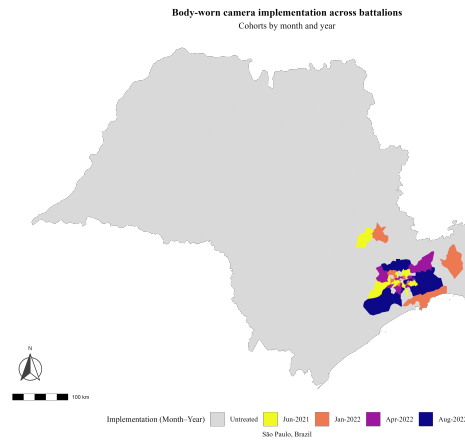


Figure 2: General map of implementation by battalion.

5.2. Police activity

This study leverages confidential administrative data provided by the São Paulo Military Police, covering the period from June 2020 to October 2023. The dataset is not publicly available due to legal and operational sensitivity, but it offers highly granular information on both police activity and criminal incidents across the state of São Paulo. Police activity is proxied by the number of recorded police confrontations, which encompass both proactive and reactive actions registered by officers. These confrontations correspond to police interventions that resulted in at least one of the following outcomes: imprisonment, injury, or death. The dataset contains geolocalized information on each such police confrontation. By comparing these outcomes across treated (BWC-equipped) and untreated battalions over time, we assess potential signs of *de-policing*, empirically observed as a reduction in the frequency of recorded police confrontations over time. Figure 3 illustrates the temporal evolution of police confrontations in the state of São Paulo.

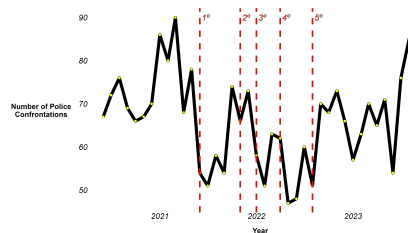


Figure 3: Time series of police confrontations. The dotted red lines represent different BWCs adoption periods.

5.3. Property crime

To capture broader public safety outcomes, we also analyze data on reports of stolen mobile phones and vehicles, two high-frequency property crimes that are widely reported and consistently recorded across space and time. These offenses are particularly suitable for empirical analysis because they rely less on discretionary police enforcement and more on victim reporting, providing a complementary measure of public safety outcomes. All crime and police data are geolocated using latitude and longitude coordinates, enabling spatial matching to battalion boundaries and facilitating a difference-in-differences design with spatially aggregated outcomes.

The data on stolen vehicles and mobile phones cover the period from 2020 to 2023, consistent with the rest of the dataset. Individual crime events are spatially assigned to approximated police battalion boundaries, which allows for the attribution of each crime to a specific battalion. Monthly counts of each type of theft are aggregated at the battalion level and aligned with the timing of body-worn camera implementation.

Map 4a shows the spatial distribution of property crimes in the state of São Paulo. Figures 4b and 4c illustrate the temporal evolution of key public safety indicators, highlighting trends in cellphone and vehicle thefts over time. These time series help visualize whether meaningful shifts occurred around the dates of BWC implementation, providing an intuitive benchmark for assessing timing and directionality prior to formal econometric modeling.

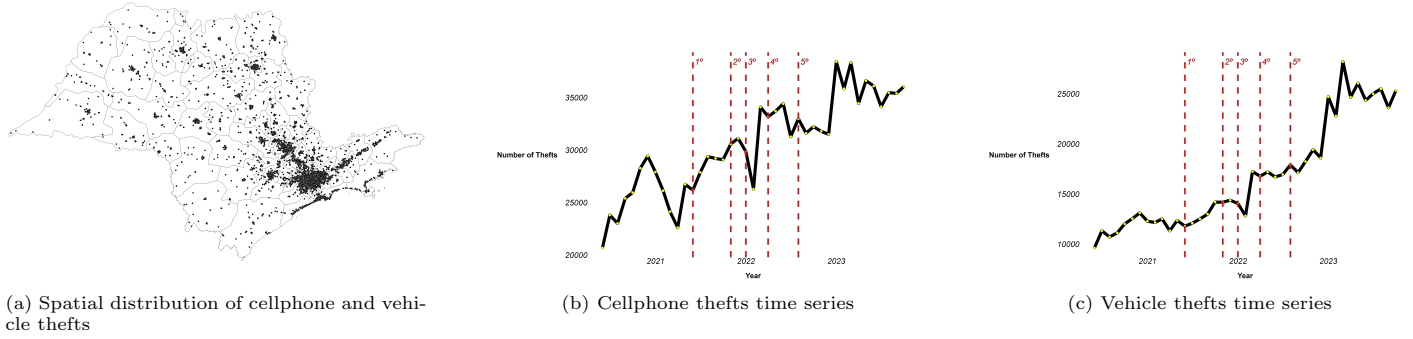


Figure 4: Maps and time series of thefts. The dotted red lines represent different BWCs adoption cohorts.

6. Methodology

We employ a quasi-experimental research design based on a difference-in-differences (DiD) framework to identify the causal effects of BWC deployment on police behavior and public safety outcomes. Our empirical strategy proceeds in three steps. First, we estimate the impact of BWCs on police confrontations as a measure of de-policing. Second, we examine whether changes in police activity translate into effects on property crime, focusing on cellphone and vehicle thefts. Third, we explore heterogeneous effects of BWCs across socioeconomic contexts and by time of day.

To empirically assess potential *de-policing* effects following the introduction of body-worn cameras, we estimate the average treatment effect on the treated (ATT) using the methodology proposed by Callaway & Sant’Anna (2021), implemented through the Callaway & Sant’Anna (2021) package. This approach allows for staggered treatment timing and incorporates doubly robust estimators, increasing credibility in settings with heterogeneous treatment effects.

Our primary outcome in this step is the number of recorded police confrontations. The data are organized as a balanced panel at the battalion-month level, aggregating all confrontations recorded within each police battalion in each month. We estimate the ATT by comparing battalions that adopted BWCs to battalions that were never treated during the study period, using dynamic event-study specifications to trace the evolution of effects over time. This analysis is first conducted on the full sample to identify the average impacts of BWCs on police activity.

Formally, the ATT is defined as:

$$ATT_{g,t} = \mathbb{E}[Y_{g,t}(1) - Y_{g,t}(0) \mid G = g],$$

where g indexes the group of battalions first treated in period g , t denotes time, $Y_{g,t}(1)$ is the potential outcome under treatment, $Y_{g,t}(0)$ is the potential outcome in the absence of treatment, and $G = g$ indicates membership in treatment cohort g .

To examine whether changes in police activity induced by BWCs have broader public safety consequences, we next analyze the effects of BWC adoption on property crime. Specifically, we focus on reported thefts of mobile phones and vehicles, two high-frequency and spatially traceable crimes in São Paulo. As in the de-policing analysis, we employ a difference-in-differences design following Callaway & Sant’Anna (2021), using a balanced panel of battalion-month observations. The estimation controls for time and unit fixed effects and exploits variation in the timing of BWC adoption across battalions, isolating the effect of BWCs from common shocks and time-invariant differences across locations.

6.1. Heterogeneous Effects by Income and Time of Day

Finally, we investigate whether the effects of BWCs on police activity and crime vary across socioeconomic contexts and by time of day. This heterogeneity analysis is motivated by both the theoretical framework and prior empirical evidence suggesting that policing incentives, risks, and visibility differ substantially across neighborhoods and temporal settings.

To capture socioeconomic heterogeneity, we use data from the 2010 Brazilian Census to construct income measures for the locations where police confrontations and crimes occurred. For each recorded event, we generate a 250-meter buffer and compute the average income of all census tracts intersecting that buffer. We then classify battalion-level outcomes into higher- and lower-income groups based on whether average income falls above or below the median across all observations.

In addition, we explore temporal heterogeneity by distinguishing between daytime (06:00–18:00) and nighttime (18:00–06:00) periods. Prior research indicates that police officers face greater discretion and reduced visibility during night shifts (Crawford & Burns, 2008), while property crimes such as cellphone and vehicle theft are more prevalent during nighttime hours. By estimating separate models by time of day, we assess whether BWCs have differential effects on police behavior and crime under conditions of lower visibility and higher risk.

In the next section, we present the empirical results of this research design. We first examine whether the adoption of body-worn cameras led to a reduction in police confrontations, providing evidence consistent with de-policing (H1). We then assess whether changes in police activity are associated with subsequent changes in property crime, focusing on cellphone and vehicle thefts (H3). Finally, we explore heterogeneity in these effects across income levels and by time of day, as predicted by the theoretical framework (H2).

7. Results

Table 2 reports ATT estimates across all outcomes, subgroups, and specifications. We discuss the findings in four steps: the baseline effect of BWCs on police confrontations, heterogeneity by neighborhood income, downstream effects on property crime, and temporal and spatial heterogeneity in crime outcomes.

7.1. BWCs and Police Activity

Police confrontations decline significantly following BWC adoption. The baseline ATT is -0.709 confrontations per battalion-month (99% CI: $[-1.211, -0.207]$). Allowing for pre-adoption behavioral adjustment yields an estimate of -0.497 (99% CI: $[-0.857, -0.137]$), with no change in sign or statistical significance. The pseudo-logarithmic specification, which down-weights high-activity outlier battalions, produces an estimate of -0.238 (99% CI: $[-0.390, -0.085]$).

Given a pre-treatment mean of roughly one confrontation per battalion-month, these estimates correspond to a reduction of approximately 20–25 percent. This magnitude is consistent with the declines in stop-and-search documented by Magaloni, Melo, & Robles (2023) in Rio de Janeiro and with the reduction in deaths from police intervention reported by Tavares, Marcolino, & Cabral (2024) in São Paulo. Event-study estimates reveal no significant pre-trends and show effects that grow in magnitude after adoption, ruling out a purely transitory response.

7.2. Heterogeneous Effects by Neighborhood Income

The aggregate decline masks substantial geographic heterogeneity. In lower-income areas, confrontation ATTs are -0.459 without anticipation (99% CI: $[-0.767, -0.151]$), -0.294 with anticipation (95% CI: $[-0.579, -0.009]$), and -0.172 in the pseudo-log specification (95% CI: $[-0.319, -0.025]$). All estimates are statistically significant.

In higher-income areas, the estimates are smaller in magnitude and statistically insignificant, with confidence intervals that include zero.

This asymmetry is the empirical centerpiece of the paper. Prior to BWC adoption, accountability in lower-income areas was limited: the detection probability for misconduct ($p^L d$) and the expected punishment cost ($p^L p \cdot C_N^L$) were both low. Cameras therefore increased the marginal cost of misconduct most sharply where oversight had previously been weakest. In higher-income areas, pre-existing oversight pressures had already constrained officer behavior, leaving little room for BWCs to shift the equilibrium.

Importantly, the higher cost of misconduct does not redirect officers toward technical engagement. In the absence of performance-based incentives for procedurally correct conduct, the dominant behavioral substitution runs from misconduct to inaction. This response selectively withdraws active policing from the communities most reliant on state security provision.

7.3. Downstream Effects on Property Crime

Vehicle theft. Vehicle theft estimates are the most consistent in the study. Level ATTs are 72.1 additional thefts per battalion-month without anticipation (99% CI: $[54.2, 89.9]$) and 82.8 with anticipation (99% CI: $[63.4, 102.1]$), both highly significant. The inverse-probability-weighted estimator yields a more conservative but still significant estimate of 13.8 (99% CI: $[10.8, 16.8]$).

The pseudo-log ATT is -0.040 and statistically insignificant. This suggests that the level increases are concentrated in larger, higher-activity battalions, where reductions in policing intensity generate the largest absolute changes in the local deterrence environment.

Cellphone theft.. Results are directionally consistent but less stable across specifications. The no-anticipation level estimate of 41.7 (99% CI: $[-2.0, 85.4]$) does not reach statistical significance, while the anticipation-inclusive model yields 83.1 (99% CI: $[22.6, 143.5]$), significant at the 99 percent level. The weighted and pseudo-log estimates are positive but imprecisely estimated.

The greater sensitivity to specification likely reflects higher cross-battalion variance in cellphone theft relative to vehicle theft.

Taken together, these results support Hypothesis 3. The pseudo-log null for both outcomes disciplines the interpretation: the public safety costs of de-policing appear to operate primarily through increases in the absolute number of crimes in high-activity jurisdictions, rather than through proportional increases across the entire battalion distribution.

Outcome	Subgroup	Method	Antic.	ATT	99% CI	Sig. 99%	Sig. 95%
Panel A: Police Activity							
Police Confrontation	Overall	Level	No	-0.709	[-1.211, -0.207]	✓	✓
		Level	Yes	-0.497	[-0.857, -0.137]	✓	✓
		Pseudo Log	Yes	-0.238	[-0.390, -0.085]	✓	✓
	Low-Income	Level	No	-0.459	[-0.767, -0.151]	✓	✓
		Level	Yes	-0.294	[-0.579, -0.009]		✓
		Pseudo Log	Yes	-0.172	[-0.319, -0.025]		✓
	High-Income	Level	No	-0.203	[-0.424, 0.018]		
		Level	Yes	-0.203	[-0.410, 0.005]		
		Pseudo Log	Yes	-0.110	[-0.232, 0.011]		
Panel B: Theft Outcomes (All Day)							
Vehicle Thefts	Overall	Level	No	72.084	[54.225, 89.944]	✓	✓
		Level	Yes	82.780	[63.435, 102.124]	✓	✓
		Pseudo Log	Yes	-0.040	[-0.122, 0.042]		
		Weighted	Yes	13.809	[10.793, 16.824]	✓	✓
Cellphone Thefts	Overall	Level	No	41.676	[-2.019, 85.372]		
		Level	Yes	83.063	[22.609, 143.518]	✓	✓
		Pseudo Log	Yes	-0.052	[-0.145, 0.042]		
		Weighted	Yes	14.104	[-5.003, 33.211]		
Panel C: Crime by Time of Day							
Vehicle Thefts	Daytime	Level	No	35.889	[26.923, 44.856]	✓	✓
		Level	Yes	44.779	[34.450, 55.108]	✓	✓
	Nighttime	Level	No	0.811	[-0.612, 2.234]		
		Level	Yes	0.749	[-0.937, 2.436]		
Cellphone Thefts	Daytime	Level	No	51.583	[25.696, 77.471]	✓	✓
		Level	Yes	59.424	[38.727, 80.121]	✓	✓
	Nighttime	Level	No	41.579	[16.438, 66.720]	✓	✓
		Level	Yes	44.744	[20.826, 68.662]	✓	✓
		Pseudo Log	Yes	0.598	[0.337, 0.860]	✓	✓
Panel D: Crime by Income Area							
Vehicle Thefts	Low-Income	Level	No	39.481	[22.640, 56.322]	✓	✓
		Level	Yes	43.438	[23.980, 62.896]	✓	✓
		Weighted	Yes	6.931	[3.355, 10.507]		✓
	High-Income	Level	No	32.609	[18.028, 47.190]	✓	✓
		Level	Yes	40.265	[22.359, 58.171]	✓	✓
		Weighted	Yes	7.162	[3.326, 10.999]	✓	✓
Cellphone Thefts	Low-Income	Level	No	3.622	[-15.716, 22.960]		
		Level	Yes	28.235	[13.113, 43.358]	✓	✓
	High-Income	Level	No	40.279	[-5.043, 85.601]		
		Level	Yes	60.030	[-11.469, 131.529]		

Note: ATT estimated following Callaway & Sant'Anna (2021). Confidence intervals are bootstrap-based. ✓ indicates the ATT is statistically significant at the corresponding level. Pseudo Log refers to the pseudo-logarithmic transformation $\log(1 + y)$. Weighted refers to the inverse-probability-weighted estimator. Anticipation (Antic.) allows for behavioral adjustment one period before formal BWC deployment.

Table 2: ATT Estimates and Confidence Intervals

7.4. Temporal and Spatial Heterogeneity in Crime Effects

Time of day.. Vehicle theft effects are concentrated during daytime hours. The daytime ATT is 35.9 without anticipation (99% CI: $[26.9, 44.9]$) and 44.8 with anticipation (99% CI: $[34.5, 55.1]$). Nighttime estimates remain close to zero, with

relatively tight confidence intervals that rule out economically meaningful effects.

This pattern is consistent with de-policing operating through reductions in visible patrol during hours when officer-citizen contact is most frequent and BWC-mediated risk aversion is most salient.

Cellphone theft responds significantly in both periods. Daytime ATTs range from 51.6 to 59.4, while the nighttime pseudo-log estimate of 0.598 log points (99% CI: [0.337, 0.860]) represents the largest proportional effect in the analysis. This finding is consistent with opportunistic street crime responding elastically to reductions in evening police presence, as Crawford & Burns (2008) associate with greater officer discretion during nighttime shifts.

Income area. For vehicle theft, significant increases appear in both lower-income areas (ATT between 39.5 and 43.4, significant at 99 percent) and higher-income areas (ATT between 32.6 and 40.3, also significant). This symmetry is notable given that confrontation reductions are detectable only in lower-income areas, suggesting spatial displacement of criminal activity as overall police presence contracts.

For cellphone theft, the anticipation-inclusive ATT in lower-income areas is a significant 28.2 (99% CI: [13.1, 43.4]). Estimates for higher-income areas are large but imprecise, likely reflecting attenuation from underreporting. Unlike vehicle theft—where insurance requirements create an independent incentive to report—cellphone theft reporting is more sensitive to trust in police institutions. Residents may therefore be less likely to report incidents to officers perceived as disengaged.

7.5. Summary

Three findings hold across outcomes and specifications. First, BWC adoption produces a robust and sustained decline in police confrontations. Second, this decline is geographically concentrated in lower-income areas, consistent with cameras imposing the largest accountability shock where baseline oversight was weakest. Third, de-policing is followed by significant increases in property crime, most clearly for vehicle theft and most strongly during daytime hours.

These results do not imply that BWCs are counterproductive: they clearly alter officer behavior. Rather, they indicate that, in the absence of complementary incentives for technical engagement, the rational response to enhanced surveillance may be withdrawal rather than improvement. In this setting, the communities most dependent on active state presence bear a disproportionate share of the resulting costs.

8. Conclusion

This paper set out to test three core hypotheses derived from the accountability and deterrence framework developed in Section 4. The results provide consistent support for the proposed mechanisms, while also clarifying their limits.

First, consistent with Hypothesis 1, BWC adoption led to a robust and sustained decline in police confrontations. This finding aligns with the prediction that increasing monitoring raises the expected cost of misconduct, thereby reducing the incidence of discretionary enforcement actions. The absence of pre-trends and the persistence of effects over time suggest that this response reflects a structural behavioral adjustment rather than a temporary compliance effect.

Second, in line with Hypothesis 2, the effects are highly heterogeneous across space. The reduction in confrontations is concentrated in lower-income areas, where baseline monitoring and accountability were weakest. This pattern is consistent with the model’s prediction that the marginal impact of surveillance is increasing in the initial level of oversight. In higher-income areas, where informal and formal accountability mechanisms were already stronger, BWCs do not significantly alter officer behavior, indicating diminishing returns to additional monitoring.

Third, the evidence supports Hypothesis 3: reductions in proactive policing are followed by increases in property crime. The results are strongest and most consistent for vehicle theft, with effects concentrated in high-activity contexts and during daytime hours. This pattern is consistent with a deterrence-based mechanism in which reduced police presence lowers the expected cost of crime, particularly in environments where visible patrol plays a central role in shaping offender behavior.

Taken together, the findings provide empirical support for a substitution mechanism at the core of the theoretical framework. BWCs successfully reduce misconduct, but in the absence of complementary incentives for procedurally correct engagement, officers respond by reducing overall activity rather than reallocating effort toward compliant enforcement. This behavioral margin—withdrawal rather than substitution toward “better” policing—is central to understanding the broader equilibrium effects of accountability technologies.

Importantly, these results do not imply that BWCs are ineffective or undesirable. On the contrary, they achieve their primary objective of reducing harmful police conduct. However, the evidence highlights a key limitation: when accountability is strengthened in isolation, without parallel incentives for productive engagement, it may generate unintended public safety costs. These costs are not evenly distributed, falling disproportionately on lower-income communities that are both more exposed to changes in policing intensity and more reliant on state-provided security.

8.1. Limitations

This section interprets the findings in light of the theoretical frameworks and empirical literature discussed earlier, but several limitations qualify how strongly we can attribute the observed patterns to the proposed mechanisms.

First, while the results are consistent with a *de-policing* mechanism driven by increased monitoring, the analysis cannot directly observe officer behavior. The interpretation of reduced confrontations as strategic disengagement remains indirect, inferred from aggregate outcomes rather than measured at the officer level. As such, alternative explanations—such as administrative changes, shifts in reporting practices, or reclassification of police activity—cannot be fully ruled out.

Second, not all police confrontations could be geolocated due to incomplete address information or data inconsistencies. The exclusion of these observations may introduce selection bias if missing cases are systematically related to specific types of encounters or neighborhoods. To the extent that this bias is spatially correlated, it may affect the estimated heterogeneity that is central to the paper’s theoretical contribution.

Third, the polygons used to define each battalion’s area of operation were constructed manually. Although care was taken to ensure spatial coherence, this process involved subjective decisions that may have introduced measurement error in the assignment of occurrences to battalions. While alternative specifications yield similar qualitative results, this limitation is relevant given the paper’s emphasis on spatial heterogeneity in policing responses.

Fourth, the analysis relies on crime data that may be subject to differential reporting across areas. This concern is particularly relevant for cellphone theft, where reporting behavior depends on trust in police institutions. If BWC adoption affects citizens’ willingness to report crimes—either positively or negatively—then part of the observed increase in crime may reflect changes in reporting rather than underlying incidence.

Fifth, the use of income data from the 2010 Brazilian census introduces potential measurement error in capturing current socioeconomic conditions. Given that the theoretical framework emphasizes baseline differences in accountability and oversight across areas, inaccuracies in income classification may attenuate or distort the estimated heterogeneity in treatment effects.

Finally, the temporal scope of the data limits the ability to assess long-term equilibrium adjustments. The observed decline in proactive policing may reflect an initial period of adaptation to increased monitoring, rather than a persistent behavioral shift. Without longer-run data, it is not possible to determine whether the documented *de-policing* effects represent a stable equilibrium or a transitional dynamic.

Taken together, these limitations suggest caution in interpreting the results as definitive evidence of the mechanisms proposed, even though the empirical patterns are strongly consistent with the theoretical framework.

8.2. Future research directions

The limitations above point to several directions for future research that would deepen our understanding of the mechanisms linking accountability, police behavior, and public safety.

A first priority is to directly measure officer behavior. Linking administrative data to officer-level activity—such as patrol intensity, GPS tracking, or detailed encounter records—would allow future studies to distinguish more precisely between reduced effort, strategic avoidance of high-risk interactions, and reallocation of policing activities. This would provide a more direct test of the *de-policing* mechanism proposed in this paper.

Second, future research should further investigate the role of reporting behavior in shaping observed crime patterns. Combining police records with victimization surveys or alternative data sources could help disentangle true changes in crime incidence from shifts in reporting, particularly in contexts where trust in law enforcement varies across communities.

Third, extending the temporal horizon of the analysis is essential to understanding whether the observed effects persist over time. Longitudinal studies could assess whether initial disengagement evolves into new policing equilibria, whether officers adapt to monitoring technologies, and whether institutions develop complementary mechanisms—such as training or performance incentives—that mitigate unintended consequences.

Fourth, improving the precision of spatial and socioeconomic data would strengthen the identification of heterogeneous effects. Updated census data, finer geographic units, and more accurate boundary definitions would allow for a more precise mapping between policing behavior and local context, which is central to the framework of geographically differentiated accountability.

Finally, future work should explore the interaction between BWCs and complementary policies. If the core mechanism identified in this paper is that increased accountability induces withdrawal in the absence of positive incentives, then combining BWCs with performance-based evaluation, training, or organizational reforms may alter this response. Testing these policy combinations would be critical for designing interventions that achieve both accountability and effective policing.

By addressing these questions, future research can move beyond documenting the existence of *de-policing* to identifying the conditions under which accountability technologies improve both police conduct and public safety.

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